Cumulants

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Positive Partition of Integer

$$\{i_1, i_2, i_3, \cdots, i_\nu\} \quad i_\mu \in \mathcal{N}^+ \quad \sum i_\nu = n \tag{1}$$

All partition is $P(n)$,for a particular partition $\pi \in P(n)$

$$|\pi| = \nu \quad \pi_j = i_j \tag{2}$$

Graph

$$m_n = 2^{\binom{n}{2}} \tag{3}$$

$$m_n = \sum_{k=0}^{n-1} \binom{n-1}{k} m_k c_{n-k} \tag{4}$$

$$m_n = \sum_{\pi \in P(n)} \prod_{B \in \pi} c_{|B|} \tag{5}$$

Moment and cumulant

$$M(t) = \langle e^{tX} \rangle = \sum_n \frac{t^n}{n!} \langle X^n \rangle \tag{6}$$

$$M(0) = 1 \quad m_n = \frac{\partial^n M(t)}{\partial t^n}|_{t=0} = \langle X^n \rangle \tag{7}$$

$$C(t) \equiv log \cdot M(t) = \log(M(t)) \equiv \sum_n \frac{t^n}{n!} c_n \tag{8}$$

$$C(0) = 0 \quad c_n = \frac{\partial^n M(t)}{\partial t^n}|_{t=0} \tag{9}$$

Moment and Cumulant relationship

$$(f(x) \cdot g(x))^{(n)} = \sum_{k=0}^{n} \binom{n}{k} f^k(x) \cdot g^{n-k}(x) \tag{10}$$

$$M^{(1)}(t) = \left(e^{C(t)}\right)^{(1)} = e^{C(t)} C^{(1)}(t) = M(t) C^{(1)}(t) \tag{11}$$

$$M^{(1+n)}(t) = \sum_{k=0}^{n} \binom{n}{k} M^{(k)}(t) C^{(n+1-k)}(t) \tag{12}$$

$$m_{n+1} = \sum_{k=0}^{n} \binom{n}{k} m_k c_{n+1-k} \tag{13}$$

Moment and Cumulant relationship

Theorem (Faà di Bruno formula)

Let n be positive integer and let f and g be two ,the coposite function $h \equiv g(f)$

$$h^{(n)}(x) = \sum_{\pi \in P_n} c_\pi g^{(\nu_\pi)}(y) \left[\prod_{j \in \pi} f^{(j)}(x) \right] \tag{14}$$

Taylor Expansion.

$$g(y) = \sum_j \frac{1}{j!} g^j y^j \quad f(x) = \sum_k f^k \frac{1}{k!} x^k \tag{15}$$

□

Moment and Cumulant relationship

$$h(x) = \sum_j \frac{1}{j!} g^j \left(\sum_k \frac{f^k}{k!} x^k\right)^j \tag{16}$$

$$= \sum_n \frac{1}{n!} n! \sum_{\nu=1}^{n} \frac{g^\nu}{\nu!} \left[\sum_{i_1+i_2+\cdots+i_\nu=n} \frac{f^{i_1} f^{i_2} \cdots f^{i_\nu}}{i_1! i_2! \cdots i_\nu!} \right] x^n \tag{17}$$

$$h^{(n)} = \sum_{\pi \in P(n)} c_\pi g^{|\pi|}(y) \left[\prod_{j \in \pi} f^{(j)}(x) \right] \tag{18}$$

$$c_\pi = \frac{n!}{(j_1!)_1^m (j_2!)_2^m \cdots (j_k!)_k^m m_1! m_2! \cdots m_k!} \tag{19}$$

Moment and Cumulant relationship

$$M(t) = e^{C(t)} \implies m_n(X) = \sum_{\pi \in P(n)} c_\pi \Pi_{j \in \pi} c_j \tag{20}$$

$$C(t) = \log(M(t)) \implies c_n(X) = \sum_{\pi \in P(n)} c_\pi (-1)^{|\pi|-1} (|\pi| - 1)! m_j \tag{21}$$

Properties of cumulants

Theorem (Homogeneity)

Suppose X is a random variable with an cumulant .Then for any $c \in \mathcal{R}$,cX has anc cumulant

$$k_n(cX) = c^n k_n(X) \tag{22}$$

Proof.

$$\begin{aligned} M_{cX}(t) = \langle e^{tcX} \rangle = C_X(ct) \\ \implies K_{cX}(t) = K_X(ct) \\ \implies K^n_{cX}(t) = c^n K^{(n)}_X(ct) \end{aligned}$$

$\square$

Properties of cumulants

Theorem (Semi-invariance)

$$k_n(X+b) = \begin{cases} k_n(X) + b & n = 1 \\ k_n(X) & n > 1 \end{cases} \tag{23}$$

Proof.

$$\begin{aligned} M_{cX}(t) = \langle e^{tcX} \rangle &= C_X(ct) \\ &\implies K_{cX}(t) = K_X(ct) \\ &\implies K^n_{cX}(t) = c^n K^{(n)}_X(ct) \end{aligned}$$

$\square$

Properties of cumulants

Theorem (Cumulants accumulate)

Suppose X and Y are independent variables, .Then $S = X + Y$

$$c_n(S) = c_n(X) + c_n(Y) \tag{24}$$

Proof.

$$M_n(S) = \langle e^{X+Y} \rangle = M_X(t) M_Y(t) \implies C(S) = C(X) + C(Y)$$

$\square$

Equilibrium Statistics

$$p_j = \frac{e^{-\beta e_j}}{\sum_{k=1}^{\infty} e^{-\beta u_k}} \quad Z(\beta) = \sum_{k=0}^{\infty} e^{-\beta u_k} \tag{25}$$

Momentum generation function of energy

$$M_U(t) = \langle E^{tU} \rangle = \sum_k e^{tu_k} \frac{e^{-\beta u_k}}{Z(\beta)} = \frac{Z(\beta - t)}{Z(\beta)} \tag{26}$$

Cumulent generating function of energy

$$C_U(t) = \log(M_U(t)) = \log(Z(\beta - t)) - \log(Z(\beta)) \tag{27}$$

Gell-Mann Low theorem

$$\langle\Omega|\, T\hat{\phi}_H(x_1)\cdots\hat{\phi}_H(x_n)\rangle = \frac{\langle 0|\, T\hat{\phi}_I(x_1)\hat{\phi}_I(x_2)\cdots\hat{\phi}_I(x_n)\hat{S}|0\rangle}{\langle 0|\hat{S}|0\rangle} \quad (28)$$

Continue

$$E_p = \frac{p^2}{2m} + \tilde{\Sigma}^{(D)} + \tilde{\Sigma}_p^{(F)} \tag{29}$$